

India's solar growth is now directly linked to battery storage with bidding mandates making battery storage element mandatory for all projects to address intermittency challenge of renewable power.

Akshay Hiranandani, CEO of Serentica Renewables, says India's battery storage ecosystem is still heavily dependent on imports, particularly of lithium-ion cells, critical minerals, and advanced battery technologies.

"Building a competitive storage ecosystem will require more than manufacturing capacity alone. The sector needs long-term policy visibility, stronger domestic supply chains, access to critical minerals, and accelerated deployment of grid-scale storage infrastructure," says Hiranandani.

The larger challenge is that India's energy transition is now moving from renewable capacity addition to renewable reliability.

NEW MANDATE

Starting June 1, only locally manufactured solar cells will be used in domestic rooftop and commercial and industrial solar projects. The Ministry of New and Renewable Energy (MNRE) directive says that all "net-metering" and "open access" solar projects commissioned after June 1 will have to use domestically made solar cells.

These are rooftop solar installations under the *PM Surya Ghar: Muft Bijli Yojana*, where consumers can offset electricity bills by supplying surplus power back to the grid while under open access, commercial and industrial consumers procure renewable power from project developers.

Initiatives such as the Production Linked Incentive (PLI) scheme, Approved List of Models



"Reducing import dependence will be critical for ensuring long-term energy security, supply chain stability, and cost competitiveness as India moves towards clean energy targets"

VINAY THADANI
DIRECTOR & CEO,
GREW SOLAR

and Manufacturers (ALMM), Domestic Content Requirement, and Basic Customs Duty have significantly accelerated investments.

Thadani says the policy focus is now moving upstream.

The larger objective is not just reducing imports but building a globally competitive renewable energy manufacturing ecosystem.

INDUSTRY PROGRESS

The industry is now moving beyond module manufacturing and focusing on deeper backward integration to strengthen supply chain resilience and reduce long-term dependence on imports.

Reliance, Adani, Tata Power, Waree, Avaada, Grew and Renew, among others, have entered into backward integration with focus on upstream solar value chain. Tata Power Renewable Energy Ltd plans to invest ₹6,500 crore to de-



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velop an ingot-wafer manufacturing capacity of up to 10 GW in two phases of 5 GW each.

The TERI paper emphasises that India should extend ALMM, which currently certifies module makers to the entire value chain. MNRE has already announced an ALMM for cells from June 2026. Now it is time to extend it to wafers and polysilicon in a year or two.

"In parallel, to manage the tariff impact, the cell and upstream ALMM transition should be ring-fenced to government and PSU-linked procurement, or to a defined annual allocation (for example, 25–50 GW), while open-access and competitively tendered private projects retain flexibility to procure least-cost modules during the scale-up period," it says. **BT**

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